

STT 465 – Homework 1: Probability Review

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Question 1: Conditional Probability and a Partition

$$\begin{aligned} H &= \{\text{home game}\}, & W &= \{\text{team wins}\}, \\ P(H) &= 0.60, & P(W | H) &= 0.70, & P(W | H^c) &= 0.45. \end{aligned}$$

- (a) By the definition of conditional probability, $P(W | H) = \frac{P(W \cap H)}{P(H)}$, so

$$P(W \cap H) = P(W | H) P(H).$$

- (b)

$$\begin{aligned} P(W \cap H) &= P(W | H) P(H) = (0.70)(0.60) = 0.42, \\ P(W \cap H^c) &= P(W | H^c) P(H^c) = (0.45)(1 - 0.60) = 0.18. \end{aligned}$$

- (c) Since $\{H, H^c\}$ partitions the sample space, by the law of total probability

$$P(W) = P(W | H) P(H) + P(W | H^c) P(H^c) = 0.42 + 0.18 = 0.60.$$

- (d) By Bayes' theorem,

$$P(H | W) = \frac{P(W | H) P(H)}{P(W)} = \frac{(0.70)(0.60)}{0.60} = \frac{0.42}{0.60} = 0.70.$$

- (e) Both conditional probabilities share the same numerator, $P(W \cap H)$, but they divide by different quantities:

$$P(H | W) = \frac{P(W \cap H)}{P(W)}, \quad P(W | H) = \frac{P(W \cap H)}{P(H)}.$$

They are equal only when $P(W) = P(H)$. That happens to be true here ($P(W) = P(H) = 0.60$), which is why part (d) gives 0.70, but in general $P(W) \neq P(H)$, so $P(H | W) \neq P(W | H)$.

Question 2: Independence and Dependence

$$P(A) = 0.40, \quad P(B) = 0.50, \quad P(A \cap B) = 0.20.$$

(a) $P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.20}{0.50} = 0.40.$

(b) $P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{0.20}{0.40} = 0.50.$

(c) A and B are independent. Two justifications:

- **Conditional probabilities:** $P(A | B) = 0.40 = P(A)$ and $P(B | A) = 0.50 = P(B)$. Knowing one event occurred does not change the probability of the other.
- **Product rule:** $P(A)P(B) = (0.40)(0.50) = 0.20 = P(A \cap B).$

(d) With $P(A \cap B) = 0.15$,

$$P(A | B) = \frac{0.15}{0.50} = 0.30.$$

Since $P(A | B) = 0.30 \neq 0.40 = P(A)$ (equivalently, $P(A \cap B) = 0.15 \neq 0.20 = P(A)P(B)$), A and B are **not** independent: knowing B occurred lowers the probability of A .

(e) Independence is not about $P(A \cap B)$ being small; it requires $P(A \cap B)$ to equal *exactly* $P(A)P(B)$. A small intersection can still mean dependence. In part (d), $P(A \cap B) = 0.15$ is smaller than in the original setup, yet the events are dependent because $0.15 \neq 0.20$. At the extreme, disjoint events with positive probabilities have $P(A \cap B) = 0$, but they are strongly dependent, since if one occurs the other cannot.

Question 3: Bayes' Theorem and Team Strength

$$S = \{\text{team is strong}\}, \quad D = \{\text{team wins its first three games}\},$$

$$P(S) = 0.30, \quad P(S^c) = 0.70, \quad P(D | S) = 0.80, \quad P(D | S^c) = 0.20.$$

(a) By the law of total probability,

$$\begin{aligned} P(D) &= P(D | S)P(S) + P(D | S^c)P(S^c) \\ &= (0.80)(0.30) + (0.20)(0.70) = 0.24 + 0.14 = 0.38. \end{aligned}$$

(b) By Bayes' theorem,

$$P(S | D) = \frac{P(D | S)P(S)}{P(D)} = \frac{0.24}{0.38} \approx 0.6316.$$

(c)

$$P(S^c | D) = \frac{P(D | S^c)P(S^c)}{P(D)} = \frac{0.14}{0.38} \approx 0.3684.$$

Check: $\frac{0.24}{0.38} + \frac{0.14}{0.38} = \frac{0.38}{0.38} = 1. \quad \checkmark$

- (d) $P(S | D)$ will **increase**. The posterior is proportional to prior $\times$ likelihood, and the likelihoods $P(D | S) = 0.80$ and $P(D | S^c) = 0.20$ do not change. Raising the prior on S from 0.30 to 0.60 increases the numerator $P(D | S)P(S)$ relative to the competing term $P(D | S^c)P(S^c)$, so more of the posterior probability goes to S . (For reference, the new posterior is $0.48/0.56 \approx 0.857$.)
- (e) The prior $P(S) = 0.30$ is the analyst's belief that the team is strong before any games are played, and the data (winning the first three games), through the likelihoods $P(D | S)$ and $P(D | S^c)$, update that belief to the posterior $P(S | D) \approx 0.63$.

Question 4: Random Variables and the Binomial Distribution

$$p = 0.75, \quad X_i = \begin{cases} 1, & \text{if the } i\text{th free throw is made,} \\ 0, & \text{otherwise,} \end{cases} \quad Y = \sum_{i=1}^{10} X_i \sim \text{Bin}(10, 0.75).$$

- (a) Each $X_i \sim \text{Bernoulli}(0.75)$, so

$$E[X_i] = 1 \cdot p + 0 \cdot (1 - p) = p = 0.75, \\ \text{Var}(X_i) = E[X_i^2] - (E[X_i])^2 = p - p^2 = p(1 - p) = (0.75)(0.25) = 0.1875.$$

- (b)

$$P(Y = 8) = \binom{10}{8} (0.75)^8 (0.25)^2 = 45 (0.75)^8 (0.25)^2 \approx 0.2816.$$

- (c)

$$P(Y \geq 8) = 1 - P(Y \leq 7) = P(Y = 8) + P(Y = 9) + P(Y = 10) \\ = \sum_{k=8}^{10} \binom{10}{k} (0.75)^k (0.25)^{10-k} \approx 0.2816 + 0.1877 + 0.0563 = 0.5256.$$

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# P(Y >= 8) where Y is Binomial(n = 10, p = 0.75)
1 - pbinom(7, size = 10, prob = 0.75)
#> [1] 0.5255928

# Equivalent: sum the pmf directly
sum(dbinom(8:10, size = 10, prob = 0.75))
#> [1] 0.5255928
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- (d) By linearity of expectation,

$$E[Y] = E\left[\sum_{i=1}^{10} X_i\right] = \sum_{i=1}^{10} E[X_i] = 10(0.75) = 7.5.$$

- (e) In general,

$$\text{Var}(Y) = \text{Var}\left(\sum_{i=1}^{10} X_i\right) = \sum_{i=1}^{10} \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

Because the free throws are independent, $\text{Cov}(X_i, X_j) = 0$ for all $i \neq j$, so the covariance terms drop out:

$$\text{Var}(Y) = \sum_{i=1}^{10} \text{Var}(X_i) = 10(0.1875) = 1.875.$$

Question 5: Joint Distribution, Marginals, and Covariance

X = number of three-point attempts, Y = number of three-point makes, with joint pmf

	$Y = 0$	$Y = 1$
$X = 1$	0.30	0.20
$X = 2$	0.10	0.40

(a) Two conditions:

1. Every entry is between 0 and 1. ✓
2. The entries sum to 1: $0.30 + 0.20 + 0.10 + 0.40 = 1.00$. ✓

So the table is a valid joint pmf.

(b)

$$\begin{aligned} P(X = 1) &= P(X = 1, Y = 0) + P(X = 1, Y = 1) = 0.30 + 0.20 = 0.50, \\ P(X = 2) &= P(X = 2, Y = 0) + P(X = 2, Y = 1) = 0.10 + 0.40 = 0.50. \end{aligned}$$

(c) The marginal of Y is $P(Y = 0) = 0.30 + 0.10 = 0.40$ and $P(Y = 1) = 0.20 + 0.40 = 0.60$. Then

$$\begin{aligned} E[X] &= \sum_x x P(X = x) = 1(0.50) + 2(0.50) = 1.50, \\ E[Y] &= \sum_y y P(Y = y) = 0(0.40) + 1(0.60) = 0.60. \end{aligned}$$

(d)

$$\begin{aligned} E[XY] &= \sum_x \sum_y xy p_{X,Y}(x, y) \\ &= (1)(0)(0.30) + (1)(1)(0.20) + (2)(0)(0.10) + (2)(1)(0.40) \\ &= 0 + 0.20 + 0 + 0.80 = 1.00. \end{aligned}$$

(e)

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y] = 1.00 - (1.50)(0.60) = 1.00 - 0.90 = 0.10.$$

X and Y are **not** independent. If they were independent, $E[XY] = E[X]E[Y]$ and the covariance would be 0; since $\text{Cov}(X, Y) = 0.10 \neq 0$, they must be dependent. We can also check directly: $P(X = 1, Y = 0) = 0.30$, but $P(X = 1)P(Y = 0) = (0.50)(0.40) = 0.20$. The positive covariance means X and Y tend to move together: more three-point attempts go with more three-point makes.